6 Root Locus

INTRODUCTION:

- The locus of the closed loop poles when the system gain (K) is varied from 0 to ∞ is known as root locus.
- W.R. Evans pioneered the root locus technique.
- Root locus also provides a measure of root sensitivity to the variations in the parameter under consideration.
- The root locus technique is applicable to both single and multiple loop system.
- The root locus can be used to design the damping ratio and natural frequency of a feedback system. It also determines the stability of the system.

Definitions

- The root locus technique is a graphical method of plotting the locus of roots in s-plane as a design parameter of the corresponding system is varied.

Root Locus Concept: -

Let us consider a second order system provided with unity negative feedback system as shown below –

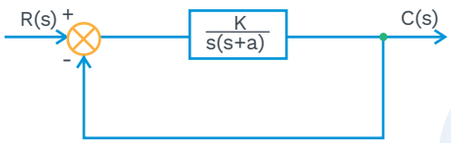


Fig 6.1 Second Order System

Gray Matter Alert!!!

- The poles of closed loop transfer function are also the roots of the characteristic equation.

Open loop transfer function (OLTF) of this system is –

$$G(s) = \frac{K}{s(s+a)}$$

Where, K represents the system gain and a represents the location of pole and both are constants

Poles of the open loop system are-

s = 0, - a

Closed loop transfer function (CLTF) of this system is –

$$CLTF = \frac{OLTF}{1+OLTF}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)}$$

Rack your Brain

- What is the use of root locus in control system?

$$= \frac{\frac{K}{s(s+a)}}{1+\frac{K}{s(s+a)}}$$

$$\frac{C(s)}{R(s)} = \frac{K}{s^2 + as + K}$$

Characteristic equation of the system is –

$$s^2 + as + K = 0$$

Roots of the equation are –

$$s_1, s_2 = -\frac{a}{2} \pm \sqrt{\left(\frac{a}{2}\right)^2 - K}$$

- If any of the system parameter (a or K) varies, the roots of the characteristic equation change.
- For positive values of a and K, the system is always stable and the roots lie in the left half of the s-plane.
- The relative stability of the system depends on the location of the roots. Let us consider the case of variable gain K while the parameter 'a' is held fixed. As K is varied from 0 to ∞, the locus of the roots s_1 and s_2 in the s-plane is explained considering three cases-

Case–1 : For $0 \le K < \frac{a^2}{4}$, the roots are distinct.

When K =0, the roots are at s_1 = 0 and s_2 = -a, which are the poles of the open loop system.

Case–2 : For $K = \frac{a^2}{4}$, the roots are real and equal i.e. $s_1 = s_2 = -\frac{a}{2}$

Case–3 : For $K \ge \frac{a^2}{4}$, the roots are complex & conjugate of each other, given by -

$$s_{1,2} = -\frac{a}{2} \pm j\frac{\sqrt{4K - a^2}}{2}$$

The real parts of the roots are fixed and the imaginary parts of the roots vary as K varies.

Example 1.

Plot the root locus for the given system $G(s) = \frac{K}{s(s+4)}$ by determining the roots of the characteristic equation.

Solution:

$$G(s) = \frac{K}{s(s+4)}$$

The characteristic equation of the system is –

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{s(s+4)} \cdot 1 = 0$$

$$s^2 + 4s + K = 0$$

Two roots is given by,

$$s_1,\ s_2 = -2 \pm \sqrt{4 - K}$$

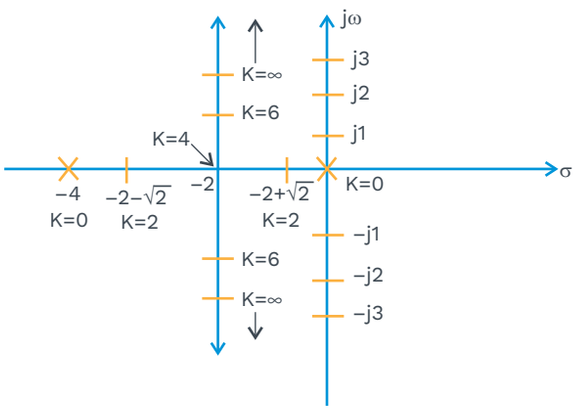
Rack your Brain

What are the advantages of the root locus technique?

For various values of K the roots are –

K	0	2	4	6	∞
s_1	0	$-2+\sqrt{2}$	–2	$-2+j\sqrt{2}$	$-2+j\infty$
s_2	–4	$-2-\sqrt{2}$	–2	$-2-j\sqrt{2}$	$-2-j\infty$

The above roots can be plotted in s-plane as –



1. For the value of K between K = 0 and K = 4, the two roots are real and negative indicating overdamped time response.

2. For K > 4, the two roots are complex with negative real part indicating that the time response is underdamped.

Angle and magnitude conditions –

For valid point on root locus, the characteristic equation of the system is given by –

$$1 + G(s)H(s) = 0$$

$$G(s)\,H(s) = -1$$

Hence, $|\,G(s)H(s)\,| = 1$.....(I) Magnitude Condition

And $\angle G(s)H(s) = \pm 180^\circ$ or $\pm (2q+1)180^\circ$.....(II) Angle condition

Equations (I) and (II) are the magnitude and angle condition respectively and the roots of the characteristic equation must satisfy the above two conditions.

- The angle condition is used for checking whether any point lie on root locus or not.
- The magnitude condition is used for finding the value of system gain (K) at any point on the root locus. It is also used for the finding the roots of the characteristic equation corresponding to a given value of the gain.
- Magnitude condition and angle condition are not independent and one implies the other.

Example – 2. Consider a system with

$$G(s)H(s) = \frac{K}{s(s+4)(s+5)}$$

Find whether s = -1 lies on root locus or not using angle condition and determine system gain (K) using magnitude condition.

Solution:

$$G(s)H(s) = \frac{K}{s(s+4)(s+5)}$$

$$G(j\omega)H(j\omega) = \frac{K}{j\omega(4+j\omega)(5+j\omega)}$$

Angle condition is –

$$\angle G(s)H(s) = \pm 180^\circ$$

$$G(s)H(s)\Big|_{s=-1} = \frac{K}{(-1)(-1+4)(-1+5)}$$

Then, $\angle G(s)H(s)\big|_{s=-1} = 0^0 - 180^\circ - 0^\circ - 0^\circ = -180^\circ$

This satisfies the angle condition. Hence, the point at s = -1 lies on the root locus.

Now, using magnitude condition –

$$\left|G(s)H(s)\right|_{s=-1} = 1$$

$$\left|\frac{K}{1\cdot\sqrt{9}\cdot\sqrt{16}}\right| = 1$$

$$K = 12$$

$\therefore$ For K = 12, one of the three roots is located at s = -1.

Example – 3. The open loop transfer function of a control system is

$$G(s)H(s) = \frac{K}{s(s+2)}$$

Determine whether s = (–1 –j) lie on the root locus or not, if so, find the value of K.

Solution:

$$G(s)H(s)\big|_{s=-1-j} = \frac{K}{(-1-j)(1-j)}$$

$$= 0° - \left(180° + 45°\right) - \left(-45°\right)$$

$$= -180^o$$

This angle condition satisfies for K > 0 and the point s = -1 -j is on the root locus.

$$|\ G(s)H(s)\ |_{s=-1-j} = 1$$

$$\left|\frac{K}{(-1-j)(1-j)}\right| = 1$$

$$\frac{K}{\sqrt{2}.\sqrt{2}} = 1$$

$$K = 2$$

Rack your Brain

Why root locus plot is always drawn on s-plane?

Construction rules for drawing root locus –

There are some following rules to draw the root locus plot –

Rule 1- The root locus is always symmetrical about the real axis.
The root of characteristic equations are either real, imaginary, complex conjugate or combination of all. Therefore, the root locus must be symmetrical about the real axis of s – plane.

- In root locus plot, closed loop poles are mirror image of each other.

Rule 2- Each branch of root locus originates at an open loop poles with K = 0 and terminates at open loop zero or at infinity with K = ∞.
Let us consider P as number of open loop poles and Z as number of open loop zeros. Then,

If P > Z then, number of branches terminating at infinity = P – Z and if Z > P then, number of branches terminating at infinity = Z – P

Consider the system having open loop transfer function as,

$$G(s)H(s) = \frac{K(s+2)}{(s+3)}$$

Then, according to rule 2, the root locus will start at s = -3 (pole) and terminate at s = -2 (zero)

Rule 3- The number of branches of the root locus will be equal to maximum. (number of poles, number of zeros).

- If the number of poles are more than number of zeros, i.e. P > Z
  Then, number of branches (N) = P
- If the number of zeros are more than number of poles, i.e. Z > P
  Then, number of branches (N) = Z
- If the number of zeros and number of poles are equal, i.e. P = Z
  Then , number of branches (N) = P = Z

For the given system –

$$G(s)H(s) = \frac{K(s+2)}{(s+3)}$$

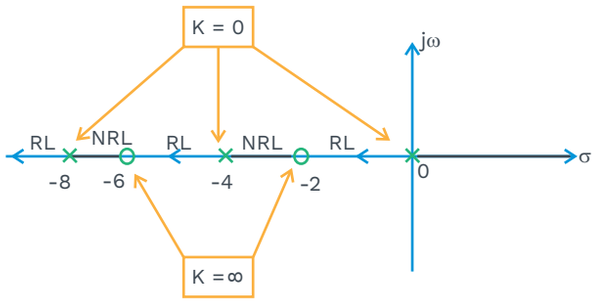
Here, P = Z = 1
So, N = 1

Rule 4- A point on the real axis of s –plane lies on the root locus if the sum of the number of open loop poles and zeros on the real axis to the right of the section is odd.

- Rule 4 has been derived from angle condition. For example, consider a system having OLTF given as –

$$G(s)H(s) = \frac{K(s+2)(s+6)}{s(s+4)(s+8)}$$

Root locus plot can be drawn as –



1. There are three open loop poles, therefore, three branches of root locus start with K = 0 at each of the poles s = 0, - 4, - 8.
2. Two branches terminate on the two open loop zeros (s = -2, -6) and one branch terminates on infinity.
3. The real axis segments between 0 and – 2, and – 4, – 6, and -8, ∞ lie on the root locus.

Rule 5- In a root locus [P-Z] branches terminate at infinity, along a set of straight lines called asymptotes, whose angles are given by –

$$\theta = \frac{(2n+1)180^\circ}{P-Z}, n = 0, 1, 2....(P-Z-1)$$

Total number of asymptotes = P – Z

For example –

$$G(s)H(s) = \frac{K}{s\left(s^2+8s+16\right)}$$

Then, P = 3

Z = 0

Number of asymptotes = P – Z

= 3 – 0 = 3

Angles of asymptotes are –

$$n = 0, \quad \theta_1 = \frac{(2\times 0+1)180^\circ}{3-0} = 60^\circ$$

$$n = 1, \quad \theta_2 = \frac{(2\times 1+1)180^\circ}{3-0} = 180^\circ$$

$$n = 2, \quad \theta_3 = \frac{(2\times 2+1)180^\circ}{3-0} = 300^\circ$$

Rule 6- Centroid of Asymptotes

The point of intersection of asymptotes with real axis is called centroid of asymptotes which is given by –

$$\sigma = \frac{\sum \text{Re(Pole)} - \sum \text{Re(Zeros)}}{P - Z}$$

Example – 4.

Consider a control system having characteristic equation $s(s+4)\left(s^2+2s+2\right)+K(s+1)=0$. Find the angle of asymptotes and the intercept of asymptotes (Centroid).

Solution:

Characteristic equation is,

$$s(s+4)\left(s^2+2s+2\right)+K(s+1)=0$$

$$1+\frac{K(s+1)}{s(s+4)\left(s^2+2s+2\right)}=0$$

Comparing the above equations with 1 + G(s)H(s) = 0

Open loop transfer function

$$G(s)H(s)=\frac{K(s+1)}{s(s+4)\left(s^2+2s+2\right)}$$

Number of poles = P = 4

$$s_p = 0, -4, \quad (-1+j1) \text{ and } (-1-j1)$$

Number of zeros = Z = 1

$$s_Z = -1$$

Number of branches terminating at infinity = P – Z

= 4 – 1 = 3

Angle of asymptotes,

$$\theta = \frac{(2n+1)180^\circ}{P-Z}$$

Put the value of n = 0, 1, 2

$$\theta = 60^\circ, 180^\circ, 300^\circ$$

$$\text{Centroid} = \frac{\sum P - \sum Z}{P-Z} = \frac{(0-4-1-1)-(-1)}{4-1} = \frac{-5}{3} = -1.67$$

Example – 5.
The characteristic equation of feedback system is given by –

$$1+K\frac{1}{s(s+1)(s+2)}=0$$

Find the value of centroid and also draw the approximate shape of root locus.

Solution:
Given,

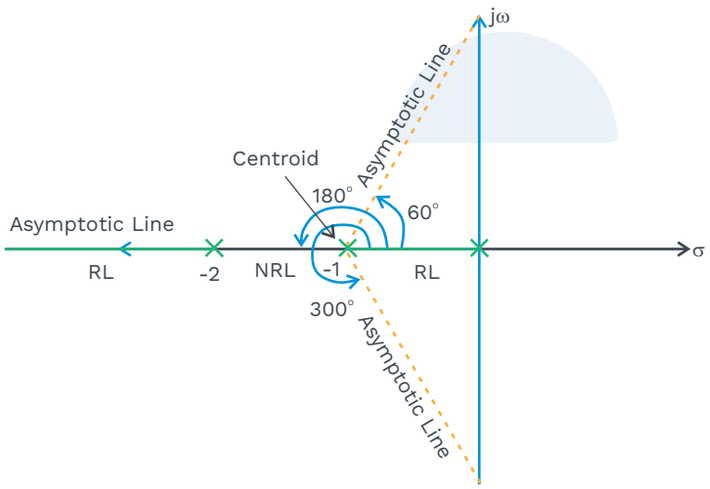
$$1+K\frac{1}{s(s+1)(s+2)}=0$$

Comparing the above expression with $1+G(s)\,H(s)=0$

$$\Rightarrow G(s)H(s)=\frac{K}{s(s+1)(s+2)}$$

Number of poles $P=3$ at $s=0,-1,-2$

Number of zeros $Z=0$



Since, there is no open loop zero, all the three branches terminate on infinite, whose angles is given by –

$$\theta=\frac{(2n+1)180^\circ}{3-0}$$

Put the value of $n=0,1,2$

$$\Rightarrow \theta=60^\circ,180^\circ,300^\circ$$

Centroid of asymptotes is given by –

$$\sigma = \frac{\sum P - \sum Z}{P - Z}$$

$$\sigma = \frac{(-1-2)-0}{3-0} = -1$$

Segments of the real axis between 0 and -1, -2 and $-\infty$ lie on the root locus plot. Approximate root locus for given system is shown above.

Rule 7- Breakaway point on real axis

- It is a point where multiple roots of the characteristic equation or multiple poles of closed loop transfer function exist.
- Each branch of root locus represents one closed loop pole.
- Breakpoints are divided into two categories –
  1. Break – away
  2. Break – in
- If break point lies between two adjacent open loop poles on real axis then it is termed as breakaway point.
- If break point lies between two adjacent open loop zeros on real axis then it is termed as break-in point.
- The breakaway or break in point can be determined by rewriting the characteristic equation and solving for the value of s from the equation given below –

$$\frac{dK}{ds} = 0$$

Thus, breakaway points are the roots of the equation $\frac{dK}{ds} = 0$. But not all the roots are actual breakaway points.

- The valid breakaway points are those roots of the equation at which the root locus angle criterion is met.

$$K = F(s)$$

Put the value of s in F(s) and find the value of K.
If $K > 0$, Valid break away point.
If $K < 0$, Invalid break away point.

Previous Year's Questions

The root locus of the feedback control system having the characteristic equation $s_2 + 6ks + 2s + 5 = 0$, where $K > 0$, enters into the real axis at–

a) $s = -1$ b) $s = -5$
c) $s = -\sqrt{5}$ d) $s = \sqrt{5}$

Answer: (b)

Example – 6.
Consider a system having open loop transfer function as

$$G(s)H(s) = \frac{K}{s\left(s^2 + 6s + 10\right)}$$

Find the value of breakaway point.

Solution:

$$G(s)H(s) = \frac{K}{s\left(s^2 + 6s + 10\right)}$$

Characteristic equation is –

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{s\left(s^2 + 6s + 10\right)} = 0$$

$$s\left(s^2 + 6s + 10\right) + K = 0$$

$$K = -s^3 - 6s^2 - 10s$$

$$\frac{dk}{ds} = -3s^2 - 12s - 10 = 0$$

$$\Rightarrow 3s^2 + 12s + 10 = 0$$

$s = -1.1835, -2.816$ are the breakaway points.

Gray Matter Alert!!!

- Break point always lies on root locus so it must satisfy angle and magnitude criterion.

Example – 7.
A control system has $G(s) = \frac{K(s+6)}{s+4}$ for a integral feedback system.
Determine the number of asymptotes and breakaway point if any.

Solution:

$$G(s) = \frac{K(s+6)}{s+4}, H(s) = \frac{1}{s}$$

$$G(s)H(s) = \frac{K(s+6)}{s(s+4)}$$

$$1 + G(s)H(s) = 0$$

Previous Year's Questions

An open loop transfer function G(s) of a system is

$$G(s) = \frac{K}{s(s+1)(s+2)}$$

For a unity feedback system, the breakaway point of the root locus on the real axis occurs at –

a) – 0.42 b)– 0.42 & - 1.58
c)– 1.58 d) None of these

Answer: (a)

$$1+\frac{K(s+6)}{s(s+4)}=0$$

$$K=\frac{-s(s+4)}{s+6}$$

For breakaway points, $\frac{dK}{ds}=0$

$$\frac{(2s+4)(s+6)-\left(s^2+4s\right)}{(s+6)^2}=0$$

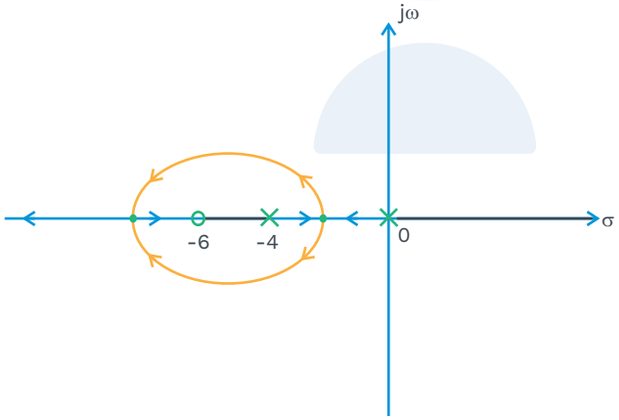
$$s^2+12s+24=0$$

$$s=-2.536,-9.464$$

$s_1 = -2.536$ is a breakaway point because it lies between two poles.
$s_2 = -9.464$ is a break-in point because it lies between two zeros [-6 and-∞]
Number of asymptotes = Number of open loop poles – Number of open loop zeros

P = 2, Z = 1

Number of asymptotes = 2 – 1 = 1



Rule 8- Angle of departure and angle of arrival –

- Angle of departure is defined as the angle at which branches of root locus depart from a complex pole.

Angle of departure from an open loop pole is given by –

$$\phi_D = 180^\circ + \phi$$

$$\phi = \phi_Z - \phi_P$$

Where,

ϕ_Z = Sum of all the angles subtended by remaining zeros.

ϕ_P = Sum of all the angles subtended by remaining poles.

- Angle of arrival is defined as the angle at which branches of root locus arrive at complex zero.

Angle of arrival at a complex zero is given by –

$$\phi_A = 180^\circ - \phi$$

$$\phi = \phi_Z - \phi_P$$

Where,

ϕ_Z = Sum of all the angles subtended by remaining zeros.

ϕ_P = Sum of all the angles subtended by remaining poles.

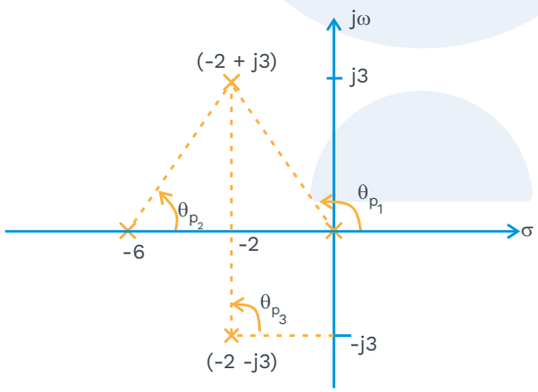
For example, the given system

$$G(s)H(s) = \frac{K}{s(s+6)\left(s^2+4s+13\right)}$$

Angle of departure is given by –

$$\phi_D = 180^\circ + \phi$$

Open loop pole zero configuration of the above system is –



$$\phi = \phi_Z - \phi_P$$

There is no zero $\therefore \phi_Z = 0$

$$\phi_P = \left(\theta_{P_1} + \theta_{P_2} + \theta_{P_3}\right)$$

$$= \left(123^\circ + 37^\circ + 90^\circ\right) = 250^\circ$$

Angle of departure is –

$$\phi_D = 180^\circ + (0 - 250^0) = -70^0$$

Example – 8.

The open loop transfer function of unity feedback system is given as –

$$G(s)H(s) = \frac{K(s+1)}{s^2 + 0.4s + 0.4} \quad 0 < K < \infty$$

Find the valid breakaway point on the real axis of s-plane and also calculate angle of departure.

Solution:

$$G(s)H(s) = \frac{k(s+1)}{s^2 + 0.4s + 0.4}$$

Characteristic equation is –

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K(s+1)}{s^2 + 0.4s + 0.4} = 0$$

$$K = \frac{-\left(s^2 + 0.4s + 0.4\right)}{(s+1)} \quad \text{.....(I)}$$

For break point:

$$\frac{dK}{ds} = 0$$

$$\frac{dK}{ds} = \frac{-\left[(2s + .4)(s+1) - \left(s^2 + 0.4s + 0.4\right)\right]}{(s+1)^2}$$

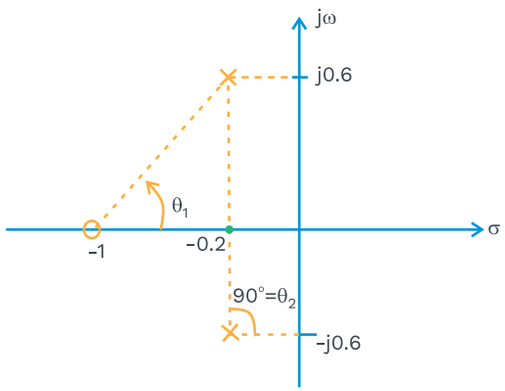
$$s^2 + 2s = 0$$

$$s = 0, -2$$

For s = –2, the value of K is positive

$\Rightarrow$ s = –2 is valid break point.

For angle of departure draw open loop pole zero graph –



Previous Year's Questions

Which one of the following statement is not correct?

a) Root locus provides insight into system stability and transient response.

b) Root locus can be used for analyzing stability and transient response.

c) Shape of root locus gives idea of type of controller needed to meet design specification.

d) Root locus can be used to handle more than one variable at a time.

Answer: (d)

$s_p = -0.2 \pm j0.6 \quad , s_z = -1$

Angle of departure is given by –

$\phi_D = 180^\circ + \phi$

$\phi = \phi_z - \phi_P$

$\phi_z = \theta_1 = \tan^{-1}\dfrac{0.6}{0.8} = 36.86$

$\phi_P = \theta_2 = 90^\circ$

$\Rightarrow \phi = 36.86 - 90^\circ = -53.13^\circ$

$\phi_D = 180^\circ + \phi$

$= 180^\circ + (-53.13^\circ)$

$\phi_D = 126.87^\circ$

Rule – 9. Intersection of root locus with imaginary axis –
The intersection of root locus branches with jω axis can be determined through Routh Hurwitz criterion.

Example – 9.

If $G(s)H(s) = \dfrac{K}{s\left(s^2 + 6s + 10\right)}$ then find the intersection of the root locus with the imaginary axis –

Solution:

Characteristic equation is –

$1 + G(s)H(s) = 0$

$1 + \dfrac{K}{s\left(s^2 + 6s + 10\right)} = 0$

$s^3 + 6s^2 + 10s + K = 0$

From Routh array table –

s^3	1	10
s^2	6	K
s^1	$\dfrac{60 - K}{6}$	
s^0	K	

We get zero row, if K = 60.
Auxiliary equation is, $A(s) = 6s^2 + K$

$$6s^2 + K = 0$$

For K = 60, the roots of the above equation lie on the jω axis and are given by –

$$s = \pm j3.16$$

Root locus branches cross the imaginary axis at s = ±j3.16 for K = 60.
Rule – 10. Intersection of asymptotes with imaginary axis
The intersection of asymptotes with the imaginary axis can be determined as for the system having OLTF is-

$$G(s)H(s) = \frac{K}{s(s+2)(s+4)}$$

$$P = 3$$

$$Z = 0$$

Number of branches terminating at infinity = 3
With the angle of $\theta = 60^\circ, 180^\circ, 300^\circ$
Characteristic Equation is –

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{s(s+2)(s+4)} = 0$$

$$K = -s(s+2)(s+4)$$

For breakaway points,

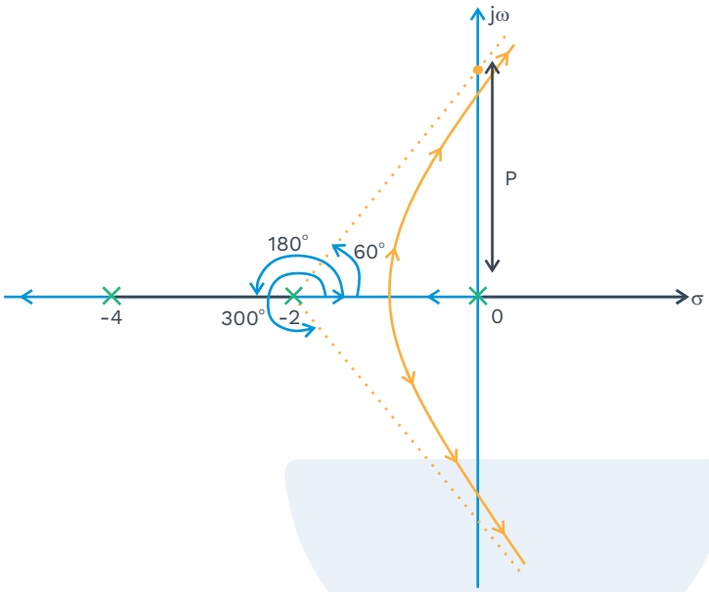
$$\frac{dK}{ds} = 3s^2 + 12s + 8 = 0$$

$$s = \frac{-12 \pm \sqrt{144 - 4 \times 24}}{6}$$

$$s = -2 \pm \sqrt{\frac{4}{3}}$$

$$s = -0.84, \quad -3.15$$

s = –3.15 is not a valid breakpoint
s = –0.84 is a valid breakpoint



$$\text{Centroid} = \frac{(-2-4)-0}{3-0} = -2$$

- Angle at which branches originate from break point

$$\Rightarrow \pm\frac{180^\circ}{n} = \pm\frac{180^\circ}{2} = \pm 90^\circ$$

- Intersection of asymptotes with imaginary axis is given by –

$$\tan\theta = \frac{P}{2}$$

$$\Rightarrow P = 2\sqrt{3}$$

Determination of K on root locus –

The value of K can be determined from the magnitude criteria –

$$|\, G(s)H(s) \,| = 1$$

Consider a system with OLTF is –

$$G(s)H(s) = \frac{K(s+3)}{s(s+1)(s+2)}$$

$$\left|\frac{K(s+3)}{s(s+1)(s+2)}\right| = 1$$

$$K = \frac{|\, s \,|\,|\, s+1 \,|\,|\, s+2 \,|}{|\, s+3 \,|}$$

$$K = \frac{|(s-0)|\,|s-(-1)|\,|s-(-2)|}{|s-(-3)|}$$

Thus, we can say that –

$$K = \frac{\text{Product of all vector lengths drawn from the poles of } G(s)H(s) \text{ to the point}}{\text{Product of all vector lengths drawn from the zeros of } G(s)H(s) \text{ to the point}}$$

Example – 9. The open loop transfer function of a feedback system is –

$$G(s)H(s) = \frac{K}{s(s+2)(s^2+6s+25)}$$

Draw the root locus for $0 \le K < \infty$.

Solution:

$$G(s)H(s) = \frac{1}{s(s+2)(s^2+6s+25)}$$

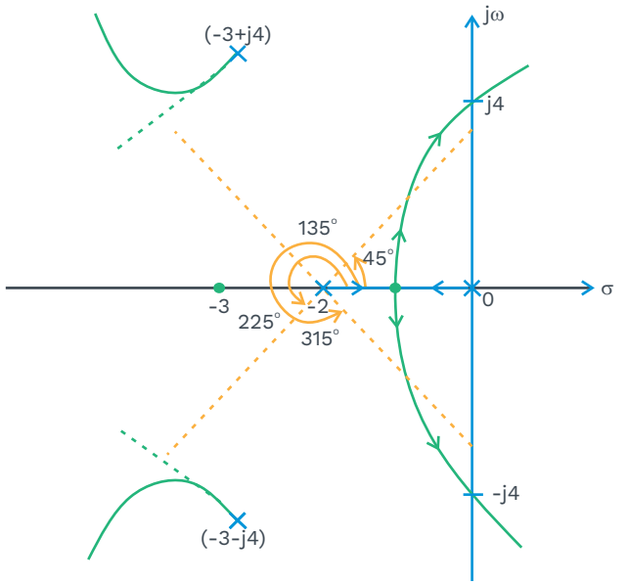
Step – 1. Poles and zeroes are –

Poles at $s_1 = 0$,

$s_2 = -2$, $s_4 = -3 - j4$

$s_3 = -3 + j4$

Step – 2. Segment on real axis between s = 0



and s = -2 is the part of the root locus

Step – 3.
Number of poles (P) = 4
Number of zero (Z) = 0
Number of root locus branches = 4
Number of asymptotes terminating at infinity = 4

Step – 4.
Centroid of asymptotes is given by –

$$\sigma = \frac{\sum P - \sum Z}{P - Z}$$

$$= \frac{(0 - 2 - 3 - 3) - (0)}{4 - 0}$$

$$= -2$$

Step – 5.

Angle of asymptotes is given by –

$$\theta = \frac{(2n + 1)180^{\circ}}{P - Z}, n = 0, 1, 2, 3$$

$$\theta_1 = \frac{(2 \times 0 + 1)180^{\circ}}{4 - 0} = 45^{\circ}$$

$$\theta_2 = \frac{(2 \times 1 + 1)180^{\circ}}{4 - 0} = 135^{\circ}$$

$$\theta_3 = \frac{(2 \times 2 + 1)180^{\circ}}{4 - 0} = 225^{\circ}$$

$$\theta_4 = \frac{(2 \times 3 + 1)180^{\circ}}{4 - 0} = 315^{\circ}$$

Step – 6.
Characteristic equation is –

$$1 + G(s)H(s) = 0$$

$$1 + \frac{K}{s(s + 2)\left(s^2 + 6s + 25\right)} = 0$$

$$K = -\left(s^4 + 8s^3 + 37s^2 + 50s\right)$$

For breakaway point –

$$\frac{dK}{ds} = 0$$

$$\frac{dK}{ds} = -\left(4s^3 + 24s^2 + 74s + 50\right)$$

$$4s^3 + 24s^2 + 74s + 50 = 0$$

$$s = -0.898, \quad -2.55, -2.55$$

Valid breakaway point = - 0.898

Step – 7
To determine intersection with imaginary axis form Routh array table –
Characteristic equation is –

$$s^4 + 8s^3 + 37s^2 + 50s + K = 0$$

Routh table is–

s^4	1	37	K
s^3	8	50	
s^2	30.75	K	
s^1	$\frac{1537.5 - 8K}{30.75}$		
s^0	K		

Auxiliary equation is –

$$30.75s^2 + K = 0$$

We get zero row, if –

$$\frac{1537.5 - 8K}{30.75} = 0$$

$$K = 192.18$$

$$\therefore 30.75s^2 = -192.18$$

$$s = \pm j2.5$$

Step – 8
Angle of departure from complex pole is –

$$\phi_D = 180^\circ + \phi$$

$$\phi = \phi_Z - \phi_P$$

$$\phi = \left(0^\circ - 104^\circ - 90^\circ - 127^\circ\right)$$

$$= -321^\circ$$

$$\phi_D = 180^\circ - 321^\circ$$

$$= -141^\circ$$

For stability range of K –

$$0 < K < 192.18$$

Closed loop system becomes unstable for

$$K > 192.18$$

Frequency of oscillation at the point of instability is –

$$K = 192.18$$

$$30.75s^2 + 192.18 = 0$$

Put $s = j\omega$ in above equation,

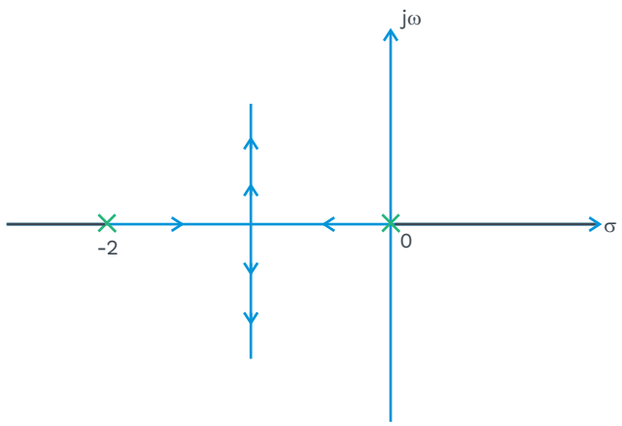
$$-30.75\omega^2 + 192.18 = 0$$

$$\omega = 2.5 \text{rad / sec}.$$

Effect of adding a pole on root locus –

Consider $G(s)H(s) = \dfrac{K}{s(s+2)}$

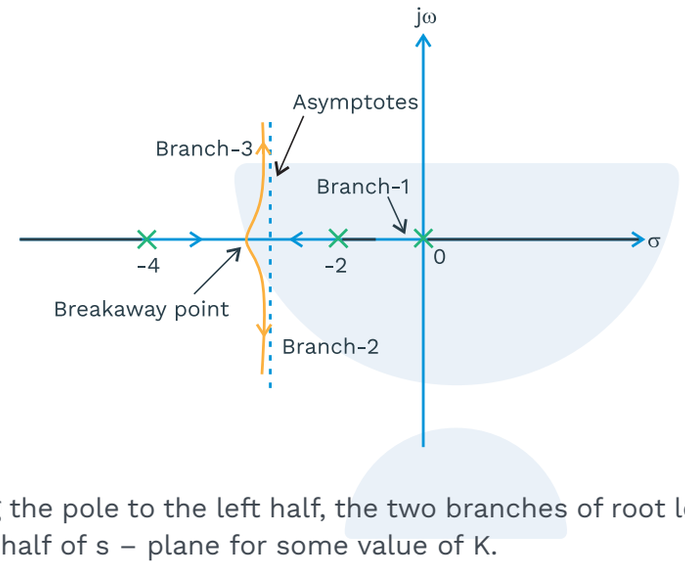
The root locus is shown below –



For any value of K the above system is stable because no pole lie in right hand plane.
Now, if we add one pole at s = -4
Then,

$$G(s)H(s) = \frac{K}{s(s+2)(s+4)}$$

The corresponding root locus can be drawn as –

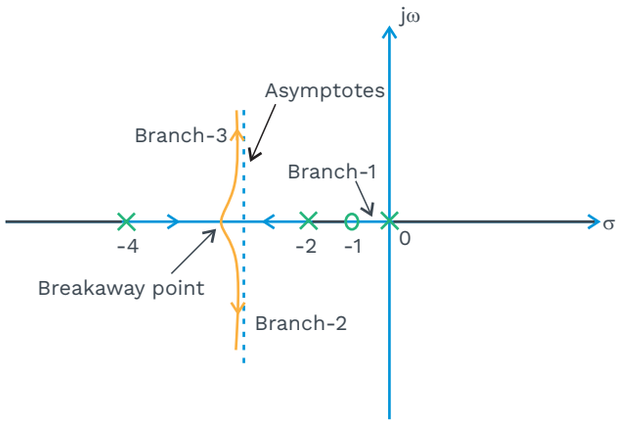


- After adding the pole to the left half, the two branches of root locus moves to the right half of s – plane for some value of K.
- This system is conditionally stable because the poles may lie in left half or right half of s-plane depending on the value of K.
  This means relative stability of closed loop system reduces.

Effect of adding a zero on root locus –
Now add zero to left half at s = -1
Then,

$$G(s)H(s) = \frac{K(s+1)}{s(s+2)(s+4)}$$

Corresponding root locus can be drawn as –



After adding a zero to the system, entire root locus plot shifts in left half of s- plane. So, the system becomes stable.
Thus, by addition of zeros the system stability increases.

Complimentary Root Locus (CRL) –
For a positive feedback closed loop control system, the characteristic equation is given by –

$$1 - G(s)H(s) = 0$$

$$G(s)H(s) = 1 + j0$$

So, magnitude condition is –

$$|G(s)H(s)| = 1$$

And angle condition is –

$$\angle G(s)H(s) = 2n\pi, n = 0, 1, 2, \ldots$$

Rules for construction of complimentary root locus (CRL)

Rule 1-CRL is symmetric about real axis.

Rule 2- CRL originates at open loop pole and terminates at open loop zero.

Rule 3- A section of real axis lies on CRL if to the right there are even number of poles and zeros.

Rule 4- Angle of asymptotes is –

$$\theta = \frac{2n\pi}{P - Z}, \text{ n = 0, 1, 2 .....upto (P-Z-1)}$$

Rule 5-

$$\text{Centroid} = \frac{\sum \text{Real}\,(\text{poles}) - \sum \text{Real}\,(\text{zeros})}{P - Z}$$

Rule 6- For breakpoint –

$$\frac{dK}{ds} = 0$$

Rule 7- Intersection with imaginary axis is same as root locus.

Rule 8- Angle of departure is given by –

$$\phi_D = 0^\circ + \phi$$

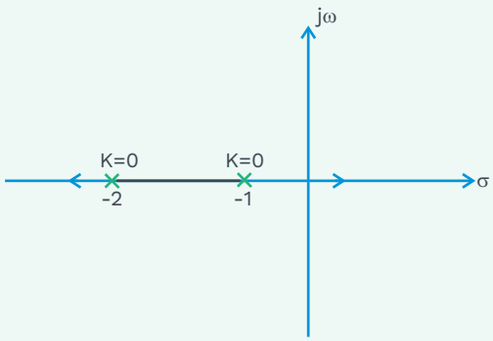
And, angle of arrival is –

$$\phi_A = 0^\circ - \phi$$

Where, $\phi = \phi_Z - \phi_P$

Previous Year's Questions

The root locus of unity feedback system is shown below in figure –



The closed loop transfer function of the system is

a) $\frac{K}{(s+1)(s+2)}$

b) $\frac{-K}{(s+1)(s+2)+K}$

c) $\frac{K}{(s+1)(s+2)-K}$

d) $\frac{K}{(s+1)(s+2)+K}$

Answer: (c)

Example 10.
Plot a root locus of the negative feedback system whose transfer function is given by –

$$G(s)H(s) = \frac{Ke^{-s}}{s(s+3)}$$

Solution:
There is transportation lag so by using the approximation of dead time,

$$G(s)H(s) = \frac{K(1-s)}{s(s+3)}$$

Or, $G(s)H(s) = \frac{-K(s-1)}{s(s+3)}$

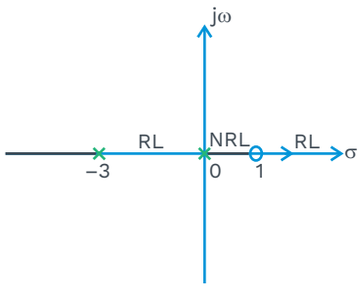
It is a positive feedback system because open loop gain factor is negative.
Rule–1. Characteristic equation is given by –

$$1 + G(s)H(s) = 0$$

$$1 - \frac{K(s-1)}{s(s+3)} = 0$$

Rule – 2. P = 2, Z = 1

Number of branch terminating at infinity = 1
Rule – 3. The section between 0 and – 3 and 1 to ∞ lies on root locus.



Rule – 4. Only one asymptote and angle is –

$$\theta = \frac{2n\pi}{P-Z} = \frac{2\times 0\times 1}{2-1}$$

$$\theta = 0^{o}$$

Rule – 5. Centroid = $\frac{\sum P - \sum Z}{P-Z}$

$$= \frac{(0-3)-(1)}{2-1} = -4$$

Rule – 6. For break point –

$$\frac{dK}{ds} = 0$$

$$K = \frac{s^2 + 3s}{s - 1}$$

$$\frac{dK}{ds} = \frac{(2s + 3)(s - 1) - \left(s^2 + 3s\right)}{(s - 1)^2} = 0$$

$$s^2 - 2s - 3 = 0$$

$$s = 3, -1$$

Both lying on CRL, so both are valid.

Gray Matter Alert!!!

- In complimentary root locus the value of gain (K) is also varied from 0 to ∞.

Rule-7. Intersection with imaginary axis –
By using Routh array –

$$s^2 + 3s - Ks + K = 0$$

s^2	1	K
s^1	3 – K	
s^0	K	

$$3 - K = 0$$

$$\Rightarrow K = 3$$

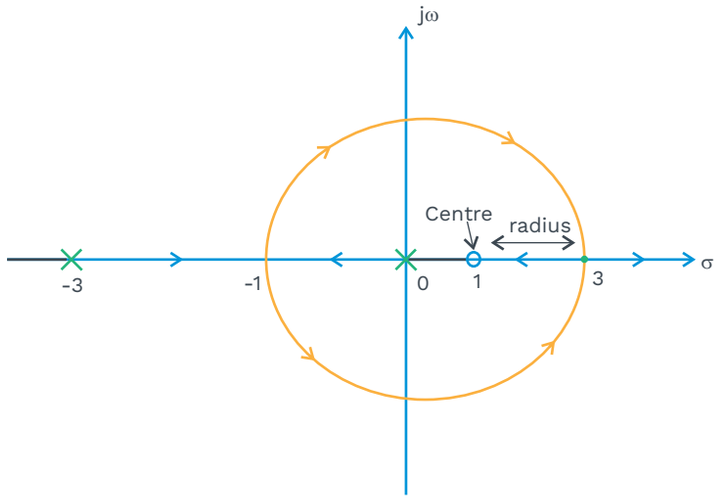
Auxiliary equation is, $A(s) = s^2 + 3 = 0$

$$s = \pm j\sqrt{3}$$

Gray Matter Alert!!!

- In CRL to check the validity of the break point we only use angle criterion.

Now, the overall CRL can be drawn as –



Example – 11.

Draw the root locus for the open loop transfer function given by –

$$G(s)H(s) = \frac{K}{s(s+4)\left(s^2+4s+13\right)}$$

And find the value of K for which the system is at the verge of instability and also find the frequency of oscillation.

Solution:

Given,

$$G(s)H(s) = \frac{K}{s(s+4)\left(s^2+4s+13\right)}$$

Step – 1. There are 4 poles at

$$s_1 = 0, \quad s_2 = -4, \quad s_3 = -2 + j3, \quad s_4 = -2 - j3$$

Step – 2. The points between 0 and – 4 lie on the root locus.

Step – 3. $P = 4$

$$Z = 0$$

Number of branches terminating at infinite = 4 – 0

$$= 4$$

Step – 4. Centroid = $\frac{(0-4-2-2)-0}{4-0} = -2$

Step – 5. Angle of asymptotes –

$$\theta = \frac{(2n+1)180^\circ}{P-Z}, n = 0,1,2,3$$

$$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$

Step – 6. Break away point is –

$$K = -s(s+4)\left(s^2+4s+13\right)$$

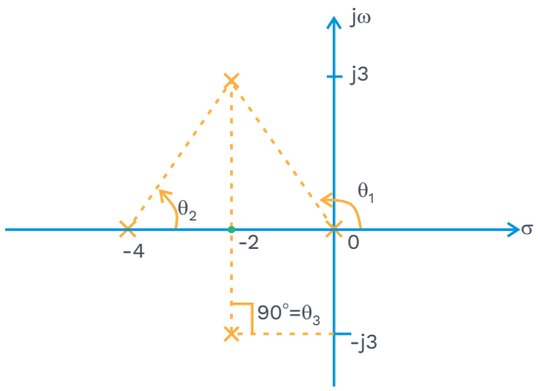
$$K = -\left(s^4+8s^3+29s^2+52s\right) = 0$$

$$\frac{dK}{ds} = -\left(4s^3+24s^2+58s+52\right) = 0$$

$$s = -2, -2 \pm j1.58$$

All break away points are valid.

Step – 7. Angle of departure is given by –



$\phi_D = 180^\circ + \phi$

$\phi = \phi_Z - \phi_P$

There is no zero $\therefore \phi_Z = 0$

$\phi_P = \theta_1 + \theta_2 + \theta_3$

$= 180^\circ - \tan^{-1}\left(\frac{3}{2}\right) + \tan^{-1}\left(\frac{3}{2}\right) + 90^\circ$

$= -270^\circ$

$\phi_D = 180^\circ + (\phi_Z - \phi_P)$

$\phi_{D_1} = 180^\circ - 270^\circ = -90^\circ$ (for pole P = –2 + j3)

$\phi_{D_2} = 90^\circ$

Step – 8. Intersection of RL with jω axis –

By using Routh array –

$s^4 + 8s^3 + 29s^2 + 52s + K = 0$

s^4	1	29	K
s^3	8	52	0
s^2	22.5	K	
s^1	$\frac{1170 - 8K}{22.5}$		
s^0	K		

For marginal stability, the value of gain is –

$$\frac{1170 - 8K}{22.5} = 0$$

For K = 146.25, Auxiliary equation is –

$$22.5s^2 + K = 0$$

$$22.5s^2 + 146.25 = 0$$

$$s = \pm j2.55$$

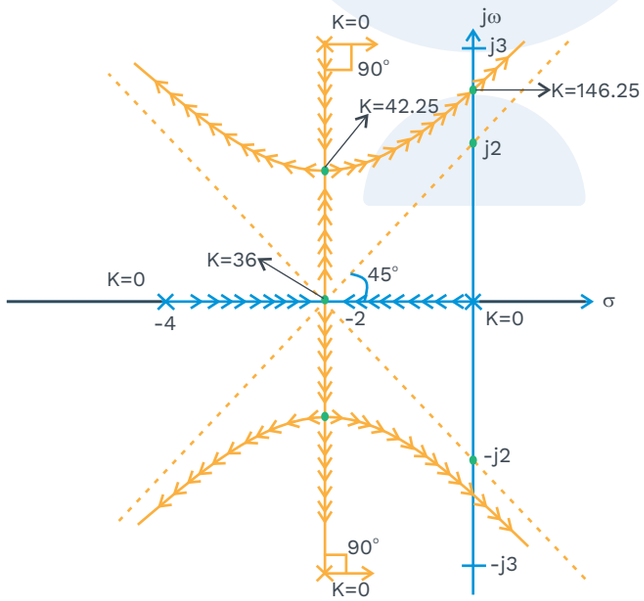
Frequency of oscillation is

$$\omega = 2.55 \text{ rad/sec.}$$

Step – 9. Intersection of asymptotes with jω axis (y) –

$$\tan 45^\circ = \frac{y}{2}, \quad y = \pm j2$$

The root locus plot is –



Example – 12.

For the transfer function $\frac{C(s)}{R(s)} = \frac{K}{s^2 + 2s + 25\alpha s + 25}$. Show that the root locus of complex roots are part of circle with centre (0, 0) and radius 5.

Solution:

Characteristic equation is

$$s^2 + 2s + 25\,\alpha s + 25 = 0$$

$$1 + \frac{25\alpha s}{s^2 + 2s + 25} = 0$$

Put $K = 25\alpha$

$$1 + \frac{Ks}{s^2 + 2s + 25} = 0$$

$$G(s)H(s) = \frac{Ks}{\left(s^2 + 2s + 25\right)}$$

Zero at s = 0

Pole at s = - 1 ± j4.89

Angle criteria for root locus –

$$\angle G(s)H(s) = \pm 180^\circ$$

$$\angle s - \angle(s + 1 + j4.89) - \angle(s + 1 - j4.89) = 180^\circ$$

Put $s = \sigma + j\omega$

$$\tan^{-1}\frac{\omega + 4.89}{\sigma + 1} + \tan^{-1}\frac{(\omega - 4.89)}{\sigma + 1} = \tan^{-1}\frac{\omega}{\sigma}$$

Taking tan on both side –

$$\frac{\frac{\omega + 4.89}{\sigma + 1} + \frac{\omega - 4.89}{\sigma + 1}}{1 - \left(\frac{\omega + 4.89}{\sigma + 1}\right)\left(\frac{\omega - 4.89}{\sigma + 1}\right)} = \frac{\omega}{\sigma}$$

$$\Rightarrow \sigma^2 + \omega^2 = 24.91$$

$$\sigma^2 + \omega^2 = 25$$

The above equation represents circle with centre at origin and radius 5.

Example – 13.

Consider a system having OLTF –

$$G(s)H(s) = \frac{K\left(s^2 - 2s + 5\right)}{(s + 2)(s - 0.5)}$$

Sketch the root locus for $0 < K < \infty$. If the gain is 0.315 then determine the value of undamped natural frequency.

Solution:

Given,

$$G(s)H(s) = \frac{K\left(s^2 - 2s + 5\right)}{(s+2)(s-0.5)}$$

Step–1 : Poles are located at s = 0.5, - 2
Zeroes are located at s = 1 + 2j, 1 – 2j

Step–2 : Segment on real axis lie between s = - 2
And s = 0.5 is the part of root locus.

Step–3 : P = 2
Z = 2
Number of roots locus branches = 2
Number of asymptote = P – Z
= 2 – 2 = 0

Step–4 : For breakaway point –

$$\frac{dK}{ds} = 0$$

$$K = -\frac{(s+2)(s-0.5)}{s^2 - 2s + 5}$$

$$\frac{dK}{ds} = \frac{-\left[(s^2 - 2s + 5)(s+2)(s-0.5) - (s+2)(s-0.5)(2s-2)\right]}{\left(s^2 - 2s + 5\right)^2}$$

$$\Rightarrow 3.5s^2 - 12s - 5.5 = 0$$

$$s = 3.838, -0.41$$

Valid breakaway point is s = - 0.41

Step–5 : Intersection with imaginary axis –

$$(K+1)s^2 + (1.5 - 2K)s + (5K - 1) = 0$$

Using Routh array –

s^2	$K+1$	$5K-1$
s^1	$1.5-2K$	
s^0	$5K-1$	

$$1.5 - 2K = 0$$

$$K = 0.75$$

Step–6 : Angle of arrival at complex pole is

$$\phi_A = 180^\circ - \phi$$

$$\phi = \phi_Z - \phi_P$$

Or $\phi_A = 180^\circ - \angle GH$

For pole s = 1 + 2j

$$G(s)H(s)\Big|_{s=1+2j} = \frac{K(s-1-2j)(s-1+2j)}{(s+2)(s-0.5)}$$

$$= \frac{K(1+2j-1+2j)}{(1+2j+2)(1+2j-0.5)}$$

$$= \frac{K(4j)}{(3+2j)(0.5+2j)}$$

$$\angle G(s)H(s)\Big|_{s=1+2j} = 90^\circ - \tan^{-1}\frac{2}{3} - \tan^{-1}\frac{2}{0.5}$$

$$= -19.645^\circ$$

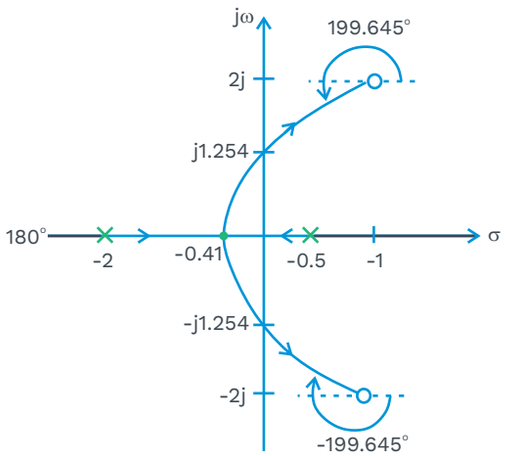
$$\phi_A = 180^\circ - (-19.645^\circ)$$

$$= 199.645^\circ$$

And at pole s = 1 -2j
Angle of arrival is same,

$$\phi_A = -199.645^\circ$$

Root locus for given system is below –



PRACTICE QUESTIONS

Q1 **The characteristic equation of a feedback control system is $s^4 + 3s^3 + 12s^2 + (K - 16)s + K = 0$ draw the root for $0 \le K < \infty$ and determine the range of K for the stability of the system.**

Sol. **(35.7 > K > 23.3)**

Q2 **A unity feedback system has an open loop transfer function –**

$$G(s) = \frac{K}{s^2(s+2)}$$

If we add a zero at s = -1 then determine the value of K for greatest damping ratio and also find the damping ratio and corresponding undamped natural frequency.

Sol. **$(K = 4,\ \zeta = 0.2,\ \omega_n = 1.76\ \text{rad / sec})$**

Q3 **The open loop transfer function of a non-unity feedback control system is given below –**

$$G(s)H(s) = \frac{K}{s(s+2)(s+4)}$$

If damping ratio (ζ) is 0.5 and variation of K is from 0 to ∞. Find the intersection point of the damping ratio line with the complex part of the root locus by analytical method.

Sol. $s = \frac{-2}{3} + \frac{j2}{\sqrt{3}}$

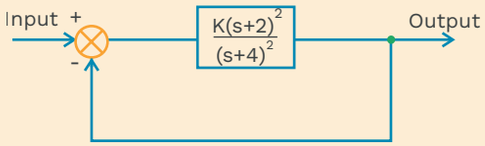
Q4 **A transfer function having OLTF –**

$$G(s)H(s) = \frac{K(s+5)(s+40)}{s^3(s+200)(s+1000)}$$

Draw the root locus plot and determine the value of centroid.

Sol. **(Centroid = - 385)**

Q5 **Draw the root locus diagram of the following system. Find the angle of departure and angle of arrival of the root locus diagram.**



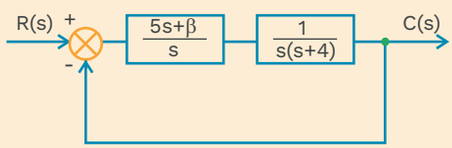
Sol. $\phi_D = \phi_A = \pm 90°$

Q6 **Given,** $G(s)H(s) = \frac{K}{s(s+1)(s+4)}$

Determine the gain margin and phase margin for the value of K such that damping ratio ζ = 0.707

Sol. **(K = 2.6, GM = 17dB, PM = 53°)**

Q7 **The block diagram of a unity feedback system is given below, draw the root locus plot as parameter β is varied from 0 to ∞. Find the value of β so that the damping ratio due to dominant closed loop pole is 0.6**



Sol. **(β = 1107)**

Q8 A unity feedback system with an open-loop transfer function is –

$$G(s)H(s) = \frac{K}{s(s+2)(s+4)}$$

Find the greatest value of K which can be used before continuous oscillations occur.

Sol. (K = 48)

Q9 Consider a OLTF $G(s) = \frac{K\left(s^2+6s+10\right)}{s^2+2s+10}$ and H(s) = 1. Draw the root locus plot on s-plane the determine the value of angle of departure is –

(a) 71.57° (b) 198.43° (c) 168.43 (d) -70.57°

Sol. Correct answer is (b).

Q10 Show that root locus for a control system with $G(s) = \frac{K\left(s^2+6s+10\right)}{s^2+2s+10}$, H(s) = 1 are the arc of circle and find the value of centre and radius of arc.

Sol. ($c = (0,0)$, $R = \sqrt{10}$)